

EVE310 Lab 9

November 4, 2025

Background

- So far in this class, we have performed an analysis (plot, create a linear regression model, etc) on one file at a time.
- Often times, there are multiple files that you would like to perform the same analysis on
- The purpose of this lab is to show you how to repeat an analysis on multiple files just by clicking “run” once

Learning objectives

- Create list of files in a directory (folder)
- Perform analysis on all csv files in a directory

Problem Statement

- Suppose you work for the UT Facilities Department and would like to analyze water consumption in each building by month
- Your boss sends you 24 data files with water consumption information
- You are asked to remove outliers and then create a boxplot by month of the water consumption in each building

Solution Approach

- First, write the code to analyze one file at a time
 - Create a boxplot by month (Lab 4)
 - Remove Outliers (Lab 7)
- Then, modify analysis to iterate over multiple files
 - Create list of files in a directory (folder)
 - Modify list to include only the files that end in .csv
 - Use list of .csv files to perform analysis

One File Solution Approach

- See Lab 9 Example Single File.ipynb

```
8 import os
9 import numpy as np
10 import matplotlib.pyplot as plt
11 import pandas as pd
12
13 #Define file name
14 file_name = 'water_DCP.csv'
15
16 #Import the data into python as a dataframe from the csv file
17 water_df = pd.read_csv(file_name)
18
19 #Based on the file name, extract the name of the building
20 building= file_name[6:9]
21
22 #THE REST OF THIS CODE IS EXACTLY THE SAME AS LABS 5 AND 7
23
24 #Convert all Date column to datetimes
25 water_df['DateTime']=pd.to_datetime(water_df['DateTime'])
26 water_df['Month']=water_df['DateTime'].dt.month
27
28 #REMOVE OUTLIERS
29
30 #Calculate the standard deviation
31 w_std = np.std(water_df['Water ( Gallons )'],ddof=1)
32
33 #Calculate the mean
34 w_mean = np.mean(water_df['Water ( Gallons )'])
35
36 #Determine what our upper threshold is
37 upper_threshold=w_mean+3*w_std
38
39 #Determine what our lower threshold is
40 lower_threshold=w_mean-3*w_std
41
42 #Only keep rows of the dataframe where the value
43 #in the Water Galons column is below the upper threshold AND above the lower threshold
44 water_df = water_df.loc[(water_df['Water ( Gallons )']<upper_threshold) &
45                          (water_df['Water ( Gallons )']>lower_threshold),:]
46
47 #CREATE BOXPLOT BY MONTH
48
49 #Make boxplot:
50 daily_boxplot = water_df.boxplot(column='Water ( Gallons )',
51                                  by='Month', grid=False, rot=45, fontsize=12)
52
53 #Set x tick mark labels to the name of the Month instead of the corresponding number:
54 plt.xticks([1,2,3,4,5,6,7,8,9,10,11,12], ['January','February','March',
55                                             'April','May','June','July','August',
56                                             'September','October','November',
57                                             'December'])
58
59 #As always, add x and y Labels, with units, and a plot title:
60 plt.ylabel('Consumption (cubic meters)', fontsize=14)
61 plt.xlabel('Month', fontsize=14)
62 plt.title(building + ' Water Consumption by Month', fontsize=14)
63
64 #Notice in the above figure that pandas added a automatic title for us.
65 #We want to remove that title and use our own, so we will use this command:
66 plt.suptitle(' ')
67
68 plt.savefig(building + '_Water_Boxplot.png',bbox_inches='tight')
69
```

One File Solution Approach

- See Lab 9 Example Single File.ipynb

```
8 import os
9 import numpy as np
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11 import pandas as pd

12
13 #Define file name
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17 water_df = pd.read_csv(file_name)
18
19 #Based on the file name, extract the name of the building
20 building= file_name[6:9]
21
22 #THE REST OF THIS CODE IS EXACTLY THE SAME AS LABS 5 AND 7
23
24 #Convert all Date column to datetimes
25 water_df['DateTime']=pd.to_datetime(water_df['DateTime'])
26 water_df['Month']=water_df['DateTime'].dt.month
```

Only this would change to
perform analysis on a
different file



REMOVE OUTLIERS

Calculate the standard deviation

```
_std = np.std(water_df['Water ( Gallons )'],ddof=1)
```

Calculate the mean

```
mean = np.mean(water_df['Water ( Gallons )'])
```

#Determine what our upper threshold is

```
upper_threshold=w_mean+3*w_std
```

#Determine what our lower threshold is

```
lower_threshold=w_mean-3*w_std
```

#Only keep rows of the dataframe where the value

#in the Water Galons column is below the upper threshold AND above the lower threshold

```
water_df = water_df.loc[(water_df['Water ( Gallons )']<upper_threshold) &
                        (water_df['Water ( Gallons )']>lower_threshold),:]
```

47 #CREATE BOXPLOT BY MONTH

48 #Make boxplot:

```
49 daily_boxplot = water_df.boxplot(column='Water ( Gallons )',
50                                  by='Month', grid=False, rot=45, fontsize=12)
```

51 #Set x tick mark labels to the name of the Month instead of the corresponding number:

```
52
53 plt.xticks([1,2,3,4,5,6,7,8,9,10,11,12], ['January','February','March',
54                                           'April','May','June','July','August',
55                                           'September','October','November',
56                                           'December'])
```

57 #As always, add x and y Labels, with units, and a plot title:

```
58
59 plt.ylabel('Consumption (cubic meters)', fontsize=14)
60 plt.xlabel('Month', fontsize=14)
61 plt.title(building + ' Water Consumption by Month', fontsize=14)
```

62 #Notice in the above figure that pandas added a automatic title for us.

63 #We want to remove that title and use our own, so we will use this command:

```
64 plt.suptitle('')
```

```
65
66 plt.savefig(building + '_Water_Boxplot.png',bbox_inches='tight')
```

Create a list of files in a directory (folder)

- Use `os.listdir()` to create a list of file names in the directory where the python script is located

```
import os

#Get a list of all the files in the folder where the script is
files=os.listdir()
```

Name	Date Modified	Size	Kind
Lab 9 Example...tiple Files.ipynb	10:54	1.7 MB	Doc
Lab 9 Example Single File.ipynb	11/28/23	76 KB	Doc
water_AHG.csv	7/19/17	54 KB	Co....csv)
water_AND.csv	7/19/17	26 KB	Co....csv)
water_ARC.csv	7/19/17	66 KB	Co....csv)
water_ART.csv	7/19/17	61 KB	Co....csv)
water_BAT.csv	7/19/17	51 KB	Co....csv)
water_BEN.csv	7/19/17	54 KB	Co....csv)
water_BHD.csv	7/19/17	64 KB	Co....csv)

```
#Get a list of all the files in the folder where the
files=os.listdir()

files

['water_CRD.csv',
 'water_BME.csv',
 'water_BMA.csv',
 'water_BAT.csv',
 'water_BLD.csv',
 'water_BUR.csv',
 'water_CDL.csv',
 'water_ARC.csv',
 'water_ART.csv',
 'water_BEN.csv',
 'water_CLA.csv',
 'water_ECj.csv',
 'water_CMA.csv',
 'water_EAS.csv',
 'water_DFA.csv',
 'water_CBA.csv',
 'water_CAL.csv',
 'water_AND.csv',
 'water_BHD.csv',
 'Lab 9 Example Single File.ipynb',
 'water_CPE.csv',
 'Lab 9 Example For Loop Multiple Files.ipynb',
 'water_EPS.csv',
 'water_AHG.csv',
 'water_DCP.csv',
 'water_CRH.csv']
```


Keep only the files that are csv files

- We do not want to try to import files that are not csv files

```
In [57]: print(files)
['Lab 13 Example For Loop Multiple Files.py', 'Lab 13 Example Single
File.py', 'Lab13.pptx', 'water_AHG.csv', 'water_AND.csv',
'water_ARC.csv', 'water_ART.csv', 'water_BAT.csv', 'water_BEN.csv',
'water_BHD.csv', 'water_BLD.csv', 'water_BMA.csv', 'water_BME.csv',
'water_BUR.csv', 'water_CAL.csv', 'water_CBA.csv', 'water_CDL.csv',
'water_CLA.csv', 'water_CMA.csv', 'water_CPE.csv', 'water_CRD.csv',
'water_CRH.csv', 'water_DCP.csv', 'water_DFA.csv', 'water_EAS.csv',
'water_ECj.csv', 'water_EPS.csv']
```

- Use a for loop to keep only the files that are csv files

```
#Create an empty list
csv_files=[]
#Loop through each of the files
for i in range(0,len(files)):
    #Check to see if '.csv' appears in the file name
    if '.csv' in files[i]:
        #If '.csv' appears in the file name, add that
        #file name to the csv_files list
        csv_files.append(files[i])
```

Keep only the files that are csv files (cont.)

- Now, the variable `csv_files` only contains the csv files that we want to analyze

```
In [59]: print(csv_files)
['water_AHG.csv', 'water_AND.csv', 'water_ARC.csv', 'water_ART.csv',
'water_BAT.csv', 'water_BEN.csv', 'water_BHD.csv', 'water_BLD.csv',
'water_BMA.csv', 'water_BME.csv', 'water_BUR.csv', 'water_CAL.csv',
'water_CBA.csv', 'water_CDL.csv', 'water_CLA.csv', 'water_CMA.csv',
'water_CPE.csv', 'water_CRD.csv', 'water_CRH.csv', 'water_DCP.csv',
'water_DFA.csv', 'water_EAS.csv', 'water_ECj.csv', 'water_EPS.csv']
```

Use for loop and csv_files list in place of file_name variable

- Now, the variable csv_files only contains the csv files that we want to analyze
- Loop through the file names in this list

```
24  #Create an empty list
25  csv_files=[]
26  #Loop through each of the files
27  for i in range(0,len(files)):
28      #Check to see if '.csv' appears in the file name
29      if '.csv' in files[i]:
30          #If '.csv' appears in the file name, add that
31          #file name to the csv_files list
32          csv_files.append(files[i])
33
34  #Loop through each of the file names and perform analysis
35  for i in range(0,len(csv_files)):
36
37      file_name=csv_files[i]
38
39      #Import the data into python as a dataframe from the csv file
40      water_df = pd.read_csv(file_name)
41
42      #Based on the file name, extract the name of the building
43      building= file_name[6:9]
44
45      #THE REST OF THIS CODE IS EXACTLY THE SAME AS LABS 5 AND 7
46
47      #Convert all Date column to datetimes
48      water_df['DateTime']=pd.to_datetime(water_df['DateTime'])
49      water_df['Month']=water_df['DateTime'].dt.month
50
51      #REMOVE OUTLIERS
```

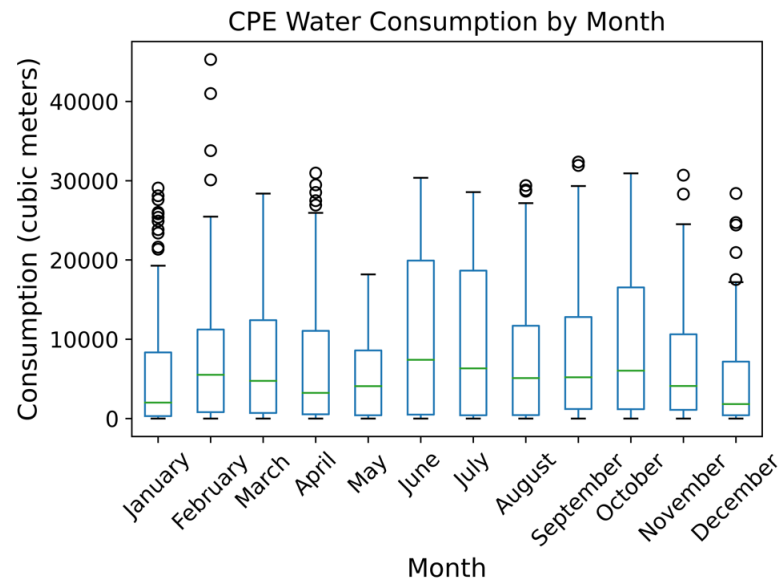
For loop, use the variable i as the index for each file name

Index csv_files to extract the file name

The rest of the analysis is the same as the single-file analysis, but inside the for loop

Results

- Multiple printed boxplots
- Only had to hit “run” once
- If it takes a while, you can go eat lunch and have your plots made when you get back!
- This example was for 24 files, but would work with many more



☐ Name	Date modified	Type	Size
AHG_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
AND_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
ARC_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
ART_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BAT_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BEN_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BHD_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BLD_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BMA_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BME_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
BUR_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CAL_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CBA_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CDL_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CLA_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CMA_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CPE_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CRD_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
CRH_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
DCP_Water_Boxplot.png	11/28/2022 1:10 AM	PNG File	
DFA_Water_Boxplot.png	11/28/2022 1:11 AM	PNG File	
EAS_Water_Boxplot.png	11/28/2022 1:11 AM	PNG File	
ECj_Water_Boxplot.png	11/28/2022 1:11 AM	PNG File	
EPS_Water_Boxplot.png	11/28/2022 1:11 AM	PNG File	

Important Caveats/Things to Remember (1)

- For Loop Variable – do not use i as another variable

```
34 #Loop through each of the file names and perform analysis
35 for i in range(0,len(csv_files)):
36
37     file_name=csv_files[i]
38
39     #Import the data into python as a dataframe from the csv file
40     water_df = pd.read_csv(file_name)
41
42     #Based on the file name, extract the name of the building
43     building= file_name[6:9]
44
45     #THE REST OF THIS CODE IS EXACTLY THE SAME AS LABS 5 AND 7
46
47     #Convert allDate column to datetimes
48     water_df['DateTime']=pd.to_datetime(water_df['DateTime'])
49     water_df['Month']=water_df['DateTime'].dt.month
50
```



- The variable i is used as the index for csv_files. If there is another for loop as part of the analysis, use a different variable as an index (j, k etc.)

Important Caveats/Things to Remember (2)

- Modify name of saved file/output based on each input file name
 - Otherwise, all saved figures will have the same file name and overwrite each other. You would be left with just 1 figure at the end

```
34 #Loop through each of the file names and perform analysis
35 for i in range(0,len(csv_files)):
36
37     file_name=csv_files[i]
38
39     #Import the data into python as a dataframe from the csv file
40     water_df = pd.read_csv(file_name)
41
42     #Based on the file name, extract the name of the building
43     building= file_name[6:9]
44
```

Building is a variable which changes based on the file being imported. Contains a 3-letter string of which building is being analyzed.

```
73
74 #Set x tick mark labels to the name of the Month instead of the corresponding
75 plt.xticks([1,2,3,4,5,6,7,8,9,10,11,12], ['January','February','March','April'])
76
77 #As always, add x and y labels, with units, and a plot title:
78 plt.ylabel('Consumption (cubic meters)', fontsize=14)
79 plt.xlabel('Month', fontsize=14)
80 plt.title(building + ' Water Consumption by Month', fontsize=14)
81
82 #Notice in the above figure that pandas added a automatic title for us.
83 #We want to remove that title and use our own, so we will use this command:
84 plt.suptitle(' ')
85
86 plt.savefig(building + '_Water_Boxplot.png',bbox_inches='tight')
87
```

The + symbol here concatenates the two strings together

Important Caveats/Things to Remember (3)

- Each csv file needs to be formatted the same way with the same type of information in each column

The image shows two side-by-side screenshots of Microsoft Excel spreadsheets. The left spreadsheet is titled 'water_BME.csv' and the right is 'water_CBA.csv'. Both spreadsheets have the same column headers: 'DateTime' in column A and 'Water (Gallons)' in column B. The data rows show dates and corresponding water volume numbers.

	DateTime	Water (Gallons)
1	2/14/2009 0:00	6600
2	2/15/2009 0:00	5600
3	2/16/2009 0:00	5900
4	2/17/2009 0:00	8300
5	2/18/2009 0:00	7800
6	2/19/2009 0:00	8900
7	2/20/2009 0:00	9900
8	2/21/2009 0:00	8800
9	2/22/2009 0:00	4900
10	2/23/2009 0:00	5800
11	2/24/2009 0:00	8000
12	2/25/2009 0:00	6300
13	2/26/2009 0:00	7600
14	2/27/2009 0:00	7100
15	2/28/2009 0:00	7800

	DateTime	Water (Gallons)
1	12/11/2010 0:00	6208
2	12/12/2010 0:00	1242
3	12/13/2010 0:00	678
4	12/14/2010 0:00	2695
5	12/15/2010 0:00	6604
6	12/16/2010 0:00	1799
7	12/17/2010 0:00	2790
8	12/18/2010 0:00	4576
9	12/19/2010 0:00	157
10	12/20/2010 0:00	141
11	12/21/2010 0:00	1387
12	12/22/2010 0:00	5479
13	12/23/2010 0:00	992
14	12/24/2010 0:00	760
15	12/25/2010 0:00	72

- In this example, each of the column headings are the same, and each column has the same type of data (first is datetime, second is a number)

Summary

- A useful technique to analyze multiple datasets at once
- Could be very useful in a future work scenario

Thanks for a great semester!